

HS202: Principles of Economics

Exam Preparation Guide

Chapter 5: Elasticity and its Application

Covers: the meaning of elasticity, price elasticity of demand and supply (formula, midpoint method, numerical examples), determinants of demand elasticity, types/degrees of elasticity (perfectly elastic to perfectly inelastic), determinants of supply elasticity, and income & cross-price elasticity of demand.

Contains: Multiple Choice Questions • Fill in the Blanks • Very Short Answer Questions • Short Answer Questions • Long Answer Questions (with key points) — all with answers/answer keys included. **Numerical problems on elasticity are a very common exam question – make sure you can apply the midpoint formula.**

Quick Revision – Key Points & Definitions

- **Elasticity** = a numerical measure of the responsiveness of Qd or Qs to one of its determinants.
- **Price elasticity of demand** = $\% \text{ change in Qd} \div \% \text{ change in P}$. It measures the price-sensitivity of buyers' demand.
- Along a demand curve, P and Q move in opposite directions (elasticity is technically negative), but by convention we **drop the minus sign and report price elasticities as positive numbers**.
- Standard $\%$ change formula = $(\text{end value} - \text{start value}) / \text{start value} \times 100\%$ — but this gives different answers depending on the direction of the change (a known problem).
- **Midpoint method** (preferred): $\% \text{ change} = (\text{end value} - \text{start value}) / \text{midpoint} \times 100\%$, where midpoint = average of start & end values. Gives the same answer regardless of direction.
- Worked example: websites price INR12,000→15,000, Qd 12→8: $\% \Delta P = 22.2\%$, $\% \Delta Q = 40.0\%$, **elasticity = $40/22.2 = 1.8$** .
- Active Learning example: hotel rooms P 700→900, Qd 5000→3000: $\% \Delta Qd = 50\%$, $\% \Delta P = 25\%$, **elasticity = $50/25 = 2.0$** .
- **Determinants of price elasticity of demand (4 examples in slides)**: (1) Availability of close substitutes → more substitutes = higher elasticity (e.g., breakfast cereal vs. water). (2) Narrowly vs. broadly defined goods → narrowly defined goods (e.g., blue jeans) are more elastic than broadly defined ones (e.g., clothing). (3) Necessities vs. luxuries → luxuries (e.g., cruises) are more elastic than necessities (e.g., insulin). (4) Time horizon → elasticity is higher in the long run than the short run (e.g., gasoline).
- **Extended list of factors affecting price elasticity of demand**: nature of the commodity (necessity/comfort/luxury), availability of substitutes, variety in uses of the commodity, postponement (can purchase be delayed), influence of habits, proportion of income spent on the commodity, range of prices, and elements of time.
- **Practical importance of price elasticity of demand**: business decisions, formulating government tax policy, determining rewards of factors of production, determining terms of trade between countries, distribution of tax burden, determination of foreign exchange rates, and decisions on nationalization of industries.
- **Five degrees/types of price elasticity of demand**: Perfectly Elastic (slight price rise → Qd falls to zero); Perfectly Inelastic (price change → no change in Qd); Unitary Elastic ($\% \Delta Qd = \% \Delta P$); Greater than Unitary/Elastic ($\% \Delta Qd > \% \Delta P$); Less than Unitary/Inelastic ($\% \Delta Qd < \% \Delta P$).
- **Graphically**: Perfectly inelastic demand = vertical D curve, elasticity = 0. Relatively steep D curve = inelastic (<1). Intermediate slope D curve = unit elastic ($=1$). Relatively flat D curve = elastic (>1). Perfectly elastic demand = horizontal D curve, elasticity = infinity.
- On a **straight-line (linear) demand curve**, the slope is constant but the **elasticity is NOT constant** — it is higher (more elastic) near the top-left (high P, low Q) and lower (more inelastic) near the bottom-right (low P, high Q).
- **Price elasticity of supply** = $\% \text{ change in Qs} \div \% \text{ change in P}$; also uses the midpoint method.
- Rule of thumb: **the flatter the supply curve, the greater/bigger the elasticity; the steeper the curve, the smaller the elasticity**.
- **Determinants of supply elasticity**: how easily sellers can change quantity produced (e.g., beachfront property is hard to vary → less elastic than new cars); and time horizon — supply elasticity is generally greater in the long run than the short run (firms can build new factories or new firms can enter).
- Supply elasticity can vary along a single supply curve — often becoming **less elastic as quantity rises**, due to capacity limits.
- **Income elasticity of demand** = $\% \text{ change in Qd} \div \% \text{ change in income}$. For **normal goods**, income elasticity > 0 ; for **inferior goods**, income elasticity < 0 .
- **Cross-price elasticity of demand** = $\% \text{ change in Qd of good 1} \div \% \text{ change in price of good 2}$. For **substitutes**, cross-price elasticity > 0 (e.g., beef price up → chicken demand up). For **complements**, cross-price elasticity < 0 (e.g., computer price up → software demand down).
- Reference textbook: N. Gregory Mankiw, *Principles of Economics*.

Section A – Multiple Choice Questions (MCQs)

1. Elasticity is best defined as:
 - (a) A measure of total revenue
 - (b) A numerical measure of the responsiveness of Q_d or Q_s to one of its determinants
 - (c) The slope of the supply curve only
 - (d) The equilibrium price
2. Price elasticity of demand is calculated as:
 - (a) $\% \text{ change in } P \div \% \text{ change in } Q_d$
 - (b) $\% \text{ change in } Q_d \div \% \text{ change in } P$
 - (c) $\text{Change in } Q_d \div \text{Change in } P$
 - (d) $Q_d \div P$
3. By convention, price elasticity of demand is reported as:
 - (a) A negative number
 - (b) A positive number (minus sign dropped)
 - (c) Always greater than 1
 - (d) Always less than 1
4. The preferred method for calculating percentage changes in elasticity, because it gives the same answer regardless of direction, is the:
 - (a) Standard method
 - (b) Midpoint method
 - (c) Average method
 - (d) Marginal method
5. In the midpoint method, the $\%$ change in a variable is calculated using:
 - (a) The end value only
 - (b) The start value only
 - (c) The midpoint (average) of start and end values as the denominator
 - (d) The government's stated value
6. In the website example, price rises from INR12,000 to 15,000 and quantity falls from 12 to 8. Using the midpoint method, the price elasticity of demand is approximately:
 - (a) 1.0
 - (b) 1.33
 - (c) 1.8
 - (d) 2.5
7. In the hotel rooms active-learning example ($P: 700 \rightarrow 900$, $Q_d: 5000 \rightarrow 3000$), the price elasticity of demand equals:
 - (a) 1.0
 - (b) 1.5
 - (c) 2.0
 - (d) 2.5
8. Which good is likely to have the HIGHER price elasticity of demand: breakfast cereal or water?
 - (a) Water, because it's essential
 - (b) Breakfast cereal, because it has close substitutes
 - (c) Both are equal
 - (d) Neither has elasticity
9. Comparing 'blue jeans' and 'clothing' in general, which has the higher price elasticity of demand?
 - (a) Clothing (broadly defined)

- (b) Blue jeans (narrowly defined)
 - (c) Both are identical
 - (d) Cannot be determined
- 10.** Comparing insulin and Caribbean cruises, which has the higher price elasticity of demand?
- (a) Insulin, because it's a necessity
 - (b) Caribbean cruises, because they are a luxury
 - (c) Both are equally elastic
 - (d) Neither is elastic
- 11.** Price elasticity of demand for gasoline is typically:
- (a) Higher in the short run than the long run
 - (b) Higher in the long run than the short run
 - (c) The same in short and long run
 - (d) Zero in all time periods
- 12.** Which of the following is NOT listed among the factors affecting price elasticity of demand?
- (a) Availability of substitutes
 - (b) Proportion of income spent on the commodity
 - (c) Influence of habits
 - (d) The color of the packaging
- 13.** A situation where a slight rise in price causes quantity demanded to fall to zero describes:
- (a) Perfectly inelastic demand
 - (b) Unitary elastic demand
 - (c) Perfectly elastic demand
 - (d) Less than unitary elastic demand
- 14.** A situation where a change in price causes no change in quantity demanded describes:
- (a) Perfectly elastic demand
 - (b) Perfectly inelastic demand
 - (c) Unitary elastic demand
 - (d) Greater than unitary elastic demand
- 15.** When the percentage change in quantity demanded equals the percentage change in price, demand is:
- (a) Perfectly elastic
 - (b) Perfectly inelastic
 - (c) Unitary elastic
 - (d) Undefined
- 16.** A perfectly inelastic demand curve is represented graphically as:
- (a) A horizontal line
 - (b) A vertical line
 - (c) A line with slope = -1
 - (d) A curve that bows outward
- 17.** A perfectly elastic demand curve is represented graphically as:
- (a) A vertical line
 - (b) A horizontal line
 - (c) A 45-degree line
 - (d) A bow-shaped curve
- 18.** On a straight-line (linear) demand curve, as you move down and to the right (lower price, higher quantity), price elasticity of demand:
- (a) Stays constant, just like the slope

- (b) Increases
- (c) Decreases (demand becomes more inelastic)
- (d) Becomes infinite

19. Price elasticity of supply is calculated as:

- (a) $\% \text{ change in } P \div \% \text{ change in } Q_s$
- (b) $\% \text{ change in } Q_s \div \% \text{ change in } P$
- (c) $Q_s \div P$
- (d) $\text{Change in } Q_s \div \text{Change in } P$

20. According to the 'rule of thumb' in the slides, a flatter supply curve indicates:

- (a) Smaller elasticity
- (b) Bigger/greater elasticity
- (c) Zero elasticity
- (d) No relationship to elasticity

21. Which is likely LESS elastic in supply: beachfront property or new cars?

- (a) New cars
- (b) Beachfront property, because its supply is harder to vary
- (c) Both are equally elastic
- (d) Neither has elasticity

22. Price elasticity of supply is generally:

- (a) Lower in the long run than the short run
- (b) Higher in the long run than the short run
- (c) The same regardless of time horizon
- (d) Always equal to 1

23. As quantity produced rises toward capacity limits, supply often becomes:

- (a) More elastic
- (b) Less elastic
- (c) Perfectly elastic
- (d) Unaffected

24. Income elasticity of demand measures:

- (a) The response of Q_d to a change in price
- (b) The response of Q_d to a change in consumer income
- (c) The response of Q_s to a change in price
- (d) The response of price to income

25. For a normal good, income elasticity of demand is:

- (a) Negative
- (b) Zero
- (c) Positive
- (d) Undefined

26. For an inferior good, income elasticity of demand is:

- (a) Positive
- (b) Negative
- (c) Always equal to 1
- (d) Undefined

27. Cross-price elasticity of demand measures:

- (a) The response of Q_d for one good to a change in the price of another good
- (b) The response of Q_d to changes in its own price
- (c) The response of Q_s to changes in price

(d) The response of income to price

28. For substitute goods, cross-price elasticity of demand is:

- (a) Negative
- (b) Positive
- (c) Zero
- (d) Always equal to 1

29. For complementary goods, cross-price elasticity of demand is:

- (a) Positive
- (b) Negative
- (c) Zero
- (d) Undefined

30. An increase in the price of beef causing an increase in demand for chicken illustrates:

- (a) Negative cross-price elasticity (complements)
- (b) Positive cross-price elasticity (substitutes)
- (c) Income elasticity
- (d) Perfectly inelastic demand

Answer Key – MCQs

1-b, 2-b, 3-b, 4-b, 5-c, 6-c, 7-c, 8-b, 9-b, 10-b, 11-b, 12-d, 13-c, 14-b, 15-c, 16-b, 17-b, 18-c, 19-b, 20-b, 21-b, 22-b, 23-b, 24-b, 25-c, 26-b, 27-a, 28-b, 29-b, 30-b

Section B – Fill in the Blanks

1. Elasticity is a numerical measure of the _____ of Qd or Qs to one of its determinants.
Answer: Responsiveness
2. Price elasticity of demand = % change in _____ ÷ % change in _____.
Answer: Qd; P
3. Price elasticities are conventionally reported as _____ numbers, even though P and Q move in opposite directions.
Answer: Positive
4. The _____ method uses the average of the start and end values as the denominator when calculating % change.
Answer: Midpoint
5. In the website price example, the price elasticity of demand was calculated to be _____.
Answer: 1.8
6. In the hotel rooms example, the price elasticity of demand was calculated to be _____.
Answer: 2.0
7. Price elasticity of demand is higher when goods have more available _____.
Answer: Substitutes
8. Price elasticity of demand is higher for _____ defined goods than for broadly defined goods.
Answer: Narrowly
9. Price elasticity of demand is higher for _____ than for necessities.
Answer: Luxuries
10. Price elasticity of demand is higher in the _____ run than in the short run.
Answer: Long
11. A situation where a slight price rise causes Qd to fall to zero is called _____ demand.
Answer: Perfectly elastic
12. A situation where a price change causes no change in Qd is called _____ demand.
Answer: Perfectly inelastic
13. When %change in Qd equals %change in P, demand is said to be _____ elastic.
Answer: Unit(ary)
14. A vertical demand curve represents _____ demand, with elasticity equal to _____.
Answer: Perfectly inelastic; 0 (zero)
15. A horizontal demand curve represents _____ demand, with elasticity equal to _____.
Answer: Perfectly elastic; infinity
16. On a linear demand curve, the slope is _____ but the elasticity _____ along the curve.
Answer: Constant; varies (changes)

17. Price elasticity of supply = % change in _____ ÷ % change in _____.

Answer: Qs; P

18. The flatter the supply curve, the _____ the elasticity of supply.

Answer: Bigger/greater

19. Supply of beachfront property is _____ elastic than supply of new cars because it is harder to vary.

Answer: Less

20. Price elasticity of supply is generally greater in the _____ run than the short run.

Answer: Long

21. Income elasticity of demand = % change in Qd ÷ % change in _____.

Answer: Income

22. For a _____ good, income elasticity of demand is positive; for a(n) _____ good, it is negative.

Answer: Normal; Inferior

23. Cross-price elasticity of demand = % change in Qd of good 1 ÷ % change in _____ of good 2.

Answer: Price

24. For substitutes, cross-price elasticity is _____; for complements, it is _____.

Answer: Positive (greater than 0); Negative (less than 0)

Section C – Very Short Answer Questions (1–2 marks each)

1. Define elasticity.

Answer: A numerical measure of the responsiveness of Qd or Qs to one of its determinants.

2. Give the formula for price elasticity of demand.

Answer: % change in quantity demanded ÷ % change in price.

3. Why do we drop the minus sign when reporting price elasticity of demand?

Answer: Because P and Q always move in opposite directions along a demand curve (making elasticity technically negative); by convention it is reported as a positive number for simplicity.

4. What is the midpoint method used for?

Answer: To calculate percentage changes in a way that gives the same answer regardless of the direction of change, using the average (midpoint) of the start and end values as the base.

5. State one determinant of price elasticity of demand.

Answer: Availability of close substitutes (more substitutes → higher elasticity) — or any of: broad/narrow definition, necessity/luxury, time horizon.

6. Define perfectly inelastic demand.

Answer: Demand where a change in price causes no change in the quantity demanded (elasticity = 0; vertical demand curve).

7. Define perfectly elastic demand.

Answer: Demand where even a slight rise in price causes quantity demanded to fall to zero (elasticity = infinity; horizontal demand curve).

8. What is unitary elastic demand?

Answer: Demand where the percentage change in quantity demanded equals the percentage change in price (elasticity = 1).

9. Give the formula for price elasticity of supply.

Answer: % change in quantity supplied ÷ % change in price.

10. State the 'rule of thumb' relating the slope of the supply curve to its elasticity.

Answer: The flatter the supply curve, the bigger the elasticity; the steeper the curve, the smaller the elasticity.

11. Why is price elasticity of supply generally greater in the long run?

Answer: Because firms have more time to build new factories, expand capacity, or new firms can enter the market.

12. Define income elasticity of demand.

Answer: The percentage change in quantity demanded divided by the percentage change in consumer income.

13. Define cross-price elasticity of demand.

Answer: The percentage change in quantity demanded of one good divided by the percentage change in the price of another good.

14. What does a positive cross-price elasticity indicate about two goods?

Answer: That the two goods are substitutes.

15. What does a negative cross-price elasticity indicate about two goods?

Answer: That the two goods are complements.

Section D – Short Answer Questions (3–5 marks each)

1. Explain the midpoint method for calculating percentage changes, and why it is preferred over the standard method. Illustrate with the website pricing example.

Answer: The standard % change formula, $(\text{end value} - \text{start value})/\text{start value} \times 100\%$, gives different answers depending on which value is treated as the 'start' — e.g., in the website example, going from A(12,000, 12) to B(15,000, 8) gives elasticity $33/25=1.33$, but going from B to A gives $50/20=2.50$. To avoid this inconsistency, the midpoint method uses the average of the start and end values as the denominator: $\% \text{ change} = (\text{end} - \text{start})/\text{midpoint} \times 100\%$. In the website example, $\% \Delta P = (15000 - 12000)/13500 = 22.2\%$, and $\% \Delta Q = (12 - 8)/10 = 40.0\%$, giving elasticity $= 40/22.2 \approx 1.8$, the same result regardless of direction.

2. Discuss, with examples, the four key determinants of price elasticity of demand covered in the slides.

Answer: (1) Availability of close substitutes: goods with many substitutes (e.g., breakfast cereal, which can be replaced by pancakes or eggs) have higher elasticity than goods with few/no substitutes (e.g., water). (2) Definition breadth: narrowly defined goods (e.g., blue jeans, with substitutes like khakis or shorts) are more elastic than broadly defined goods (e.g., clothing in general, which has very few substitutes). (3) Necessities vs luxuries: necessities (e.g., insulin) have low elasticity because people must buy them regardless of price, while luxuries (e.g., Caribbean cruises) have high elasticity because they can easily be forgone. (4) Time horizon: elasticity is higher in the long run than the short run (e.g., gasoline), because people have more time to adjust behaviour — buying smaller cars, moving closer to work, etc.

3. Explain the five degrees/types of price elasticity of demand with their graphical representation.

Answer: (1) Perfectly elastic demand: elasticity = infinity; a horizontal demand curve; even a tiny price rise drives Qd to zero. (2) Perfectly inelastic demand: elasticity = 0; a vertical demand curve; price changes cause no change in Qd. (3) Unitary elastic demand: elasticity = 1; an intermediate-slope curve; $\% \Delta Q_d$ equals $\% \Delta P$. (4) Greater than unitary (elastic) demand: elasticity > 1 ; a relatively flat curve; $\% \Delta Q_d$ exceeds $\% \Delta P$. (5) Less than unitary (inelastic) demand: elasticity < 1 ; a relatively steep curve; $\% \Delta Q_d$ is smaller than $\% \Delta P$. Note that although the slope of a straight-line demand curve is constant, its elasticity varies along the curve — more elastic toward the upper-left (high price, low quantity) and less elastic toward the lower-right.

4. Explain price elasticity of supply, its determinants, and how it can vary along a single supply curve.

Answer: Price elasticity of supply = $\% \text{ change in quantity supplied} \div \% \text{ change in price}$, measuring sellers' price-sensitivity. Like demand, it uses the midpoint method for calculation. The main determinant is how easily sellers can vary the quantity they produce — e.g., beachfront property (a fixed resource, hard to vary) has low elasticity of supply, while new cars (production can be scaled up more easily) have higher elasticity. Time horizon also matters: elasticity of supply is generally greater in the long run, since firms can build new factories or new firms can enter the market. Supply elasticity can also change along a single supply curve — for instance, supply is often relatively elastic at low quantities but becomes progressively less elastic (steeper) as output approaches capacity limits.

5. Explain income elasticity of demand and cross-price elasticity of demand, with examples and real-world illustrations (e.g., the news headlines about gas prices).

Answer: Income elasticity of demand = $\% \text{ change in } Q_d \div \% \text{ change in consumer income}$. For normal goods, income elasticity is positive (higher income $\rightarrow$ higher demand); for inferior goods, it is negative (higher income $\rightarrow$ lower demand, as consumers switch to better alternatives). Cross-price elasticity of demand = $\% \text{ change in } Q_d \text{ of one good} \div \% \text{ change in price of another good}$. For substitutes (e.g., beef and chicken), it is positive — a price rise in beef increases demand for chicken. For complements (e.g., computers and software), it is negative — a price rise in computers decreases demand for software. Real-world illustrations from the slides: rising gas prices led buyers to flock to small cars, students to shift to online courses, more bicycle sales, increased demand for camels in India (as tractor substitutes), and farmers switching to mules — all examples of positive cross-price elasticity between gasoline and its substitutes.

Section E – Long Answer Questions

These are the high-probability, exam-favourite topics for long/essay-type answers based on the instructor's slides. Expect at least one numerical elasticity calculation and one conceptual/determinants question in the exam. Each question lists the key points your answer must cover.

1. Explain the concept of price elasticity of demand in detail: definition, formula, the midpoint method (with full worked calculation), and why elasticities are reported as positive numbers.

- Define elasticity generally, then price elasticity of demand specifically.
- Give the formula: $\% \Delta Q_d / \% \Delta P$.
- Explain why the minus sign is dropped (P and Q move in opposite directions along the D curve).
- Explain the standard % change formula and its 'start point' problem with a numeric illustration ($33/25=1.33$ vs $50/20=2.50$).
- Introduce and define the midpoint method formula, and explain why it solves the start-point problem.
- Work through the full website example step by step: $\% \Delta P = (15000-12000)/13500=22.2\%$, $\% \Delta Q = (12-8)/10=40\%$, elasticity $= 40/22.2=1.8$.
- Optionally include the hotel rooms active-learning example as a second worked illustration (elasticity $= 2.0$).
- Conclude by noting that this elasticity value tells us demand is 'elastic' — buyers are quite price-sensitive.

2. Discuss in detail the determinants of price elasticity of demand, using all the examples from the slides (cereal/water, jeans/clothing, insulin/cruises, gasoline short-run/long-run), and list the extended set of factors.

- State that elasticity depends on several factors, illustrated via paired examples where prices rise by the same 20%.
- Example 1: breakfast cereal (many substitutes → more elastic) vs. water (no substitutes → less elastic). Lesson: substitutes raise elasticity.
- Example 2: blue jeans (narrow definition → more elastic) vs. clothing (broad definition → less elastic). Lesson: narrower goods are more elastic.
- Example 3: insulin (necessity → less elastic) vs. Caribbean cruises (luxury → more elastic). Lesson: luxuries are more elastic than necessities.
- Example 4: gasoline short run (less elastic, few options) vs. long run (more elastic, people can adapt — smaller cars, relocate). Lesson: elasticity rises over time.
- List the extended factors: nature of the commodity, availability of substitutes, variety in uses, postponement, habits, proportion of income spent, range of prices, elements of time.
- Briefly mention practical importance: business pricing decisions, government tax policy, factor rewards, terms of trade, tax burden distribution, exchange rate determination, nationalisation decisions.

3. Explain the different degrees/types of price elasticity of demand with graphical description of each, and explain why elasticity varies along a linear demand curve even though its slope is constant.

- Define and describe each of the five types with elasticity value and D-curve shape: Perfectly Elastic (∞ , horizontal), Greater than Unitary/Elastic (>1 , relatively flat), Unitary Elastic ($=1$, intermediate slope), Less than Unitary/Inelastic (<1 , relatively steep), Perfectly Inelastic (0 , vertical).
- Give the numeric illustration for each from the slides (e.g., $\% \Delta Q = 0\% / \% \Delta P = 10\% \rightarrow 0$; $\% \Delta Q = 10\% / \% \Delta P = 10\% \rightarrow 1$; $\% \Delta Q > 10\% / \% \Delta P = 10\% \rightarrow >1$; etc.).
- Explain the linear demand curve elasticity paradox: even though slope ($\Delta P / \Delta Q$, or its inverse) stays constant along a straight line, elasticity — which depends on percentage changes relative to the current P and Q levels — is NOT constant.

- Use the numeric example from the slides: at high P/low Q, elasticity=5.0; at midpoint, elasticity=1.0; at low P/high Q, elasticity=0.2 — showing elasticity falls as you move down the same straight demand line.
- Conclude: slope and elasticity are related but distinct concepts; never assume a straight demand line has uniform elasticity.

4. Explain price elasticity of supply in detail — definition, formula, the 'flatter curve = more elastic' rule, determinants, and how elasticity can vary along a single supply curve due to capacity constraints.

- Define price elasticity of supply and give its formula ($\% \Delta Q_s / \% \Delta P$), noting the midpoint method is used the same way as for demand.
- Work through the numeric slide example: $\% \Delta Q_s = 16\%$, $\% \Delta P = 8\%$, elasticity=2.0 (elastic supply).
- State the 'rule of thumb': flatter supply curve = bigger elasticity; steeper = smaller elasticity.
- List the five classifications of supply elasticity (mirroring demand): perfectly elastic, perfectly inelastic, unitary, less than unitary, more than unitary.
- Explain determinant 1: ease of varying quantity produced — beachfront property (fixed, low elasticity) vs. new cars (scalable, higher elasticity).
- Explain determinant 2: time horizon — supply elasticity greater in the long run (new factories/firms can enter) than short run.
- Explain how a single supply curve's elasticity can change: often becomes less elastic as quantity rises toward capacity limits (using the slide's numeric P/Q example, e.g., elasticity > 1 at low output but < 1 near capacity).

5. Discuss income elasticity of demand and cross-price elasticity of demand in detail, with their formulas, sign interpretation, and real-world/news illustrations.

- Define income elasticity of demand and its formula: $\% \Delta Q_d / \% \Delta \text{income}$.
- Explain sign interpretation: positive for normal goods (demand rises with income), negative for inferior goods (demand falls as income rises, since consumers switch away).
- Define cross-price elasticity of demand and its formula: $\% \Delta Q_d(\text{good 1}) / \% \Delta P(\text{good 2})$.
- Explain sign interpretation: positive for substitutes (e.g., beef price up → chicken demand up); negative for complements (e.g., computer price up → software demand down).
- Bring in the real-world news headlines cited in the slides as illustrations of cross-price elasticity in action: rising gas prices leading to small-car sales, online course enrolment, bicycle sales, camel demand in India (substitute for tractors), and farmers switching to mules.
- Conclude by tying elasticity concepts together: price elasticity (own price), income elasticity (income), and cross-price elasticity (related goods' prices) together explain the full set of demand determinants covered under the Law of Demand's 'other things equal' clause.